



Cambridge IGCSE™

MATHEMATICS

0580/42

Paper 4 (Extended)

October/November 2020

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

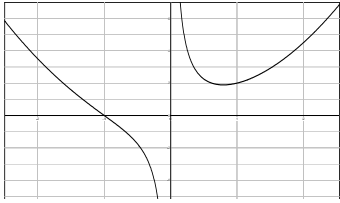
Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	9080 cao	3	B2 for 9078 to 9081... or M1 for $813 \times \text{their } 11\text{h } 10\text{min}$
1(b)(i)	654 or 653.5...	2	M1 for $10260 \div 15 \text{ h } 42 \text{ min}$ oe
1(b)(ii)(a)	21.8 or 21.82 to 21.83	1	
1(b)(ii)(b)	4.58 or 4.59 cao	2	M1 for $470 \div (10260 \div 100)$ oe or $100 \div \text{their } \mathbf{(b)(ii)(a)}$
1(c)	12.97	1	
2(a)	Translation $\begin{pmatrix} 1 \\ -6 \end{pmatrix}$	2	B1 for each
2(b)(i)	Image at (0, 1), (–3, 1), (–3, 2)	2	B1 for reflection in $x = k$ or $y = 1$
2(b)(ii)	Image at (5, –4), (5, –1), (4, –1)	2	B1 for rotation 90° anticlockwise with other centre or for rotation 90° clockwise about (6, 0)
2(b)(iii)	Image at (–1, –2), (–7, –2), (–7, –4)	2	B1 for enlargement, factor –2 with other centre
3(a)(i)	2210 or 2208 or 2208.2, or 2208.16...	2	M1 for $2000 \times \left(1 + \frac{2}{100}\right)^5$ oe
3(a)(ii)	10.4 or 10.5 or 10.40 to 10.41	2	M1 for $\frac{\text{their } \mathbf{(a)(i)} - 2000}{2000} [\times 100]$ or $\frac{\text{their } \mathbf{(a)(i)}}{2000} \times 100$ or $\left(1 + \frac{2}{100}\right)^5 - 1$ or $\left(1 + \frac{2}{100}\right)^5 \times 100$ oe
3(a)(iii)	12	3	B2 for 11.3 or 11.26 to 11.27 OR M2 for $[2000 \times] \left(1 + \frac{2}{100}\right)^{11}$ oe or $[2000 \times] \left(1 + \frac{2}{100}\right)^{12}$ oe seen or M1 for $[2000 \times] \left(1 + \frac{2}{100}\right)^n$ oe, $n > 5$ oe or for $2000 \times \left(1 + \frac{2}{100}\right)^n = \text{or } > \text{or } \geq 2500$ oe

Question	Answer	Marks	Partial Marks
3(b)	490 cao	3	M2 for $p \times \left(1 - \frac{4}{100}\right)^{16} = 255$ oe soi by 490.0... or M1 for $p \times \left(1 - \frac{4}{100}\right)^n = 255$ oe, $n > 1$ oe
4(a)(i)	25	1	
4(a)(ii)	10 nfww	2	B1 for [lq =] 22 or [uq =] 32
4(a)(iii)	27	1	
4(a)(iv)	6	2	B1 for 114 written
4(b)(i)	27.9 or 27.91 to 27.92 nfww	4	M1 for mid-values M1 for $\sum fx$ where x lies within or on boundary of correct interval M1 dep $\sum fx \div 120$ dep on second M1
4(b)(ii)	7.6	2	M1 for $\frac{18}{10}$ oe or $\frac{38}{20}$ oe or B1 for [multiplier] 4 or $\frac{1}{4}$
5(a)	1.48	3	B2 for $7x + 2 = 12.36$ or better or M1 for $3x + 2(2x + 1) [= 12.36]$ or better
5(b)	1.75 or $1\frac{3}{4}$	3	B2 for $18x - 14x = 7$ or better or M1 for $18x = 7(2x + 1)$
5(c)	[0].8 oe	3	B2 for $4(2x + 1) = 13x$ or M1 for $\frac{4}{x} = \frac{13}{2x+1}$ oe or correct equation to find number of cakes

Question	Answer	Marks	Partial Marks
5(d)	$\frac{20}{x} + \frac{10}{2x+1} = 45$ oe	M2	B1 for $\frac{20}{x}$ seen or $\frac{10}{2x+1}$ seen
	$90x^2 - 5x - 20 [= 0]$ oe	B2	B1 for $\frac{20(2x+1)+10x}{x(2x+1)} = 45$ or better
	$(9x+4)(2x-1) [= 0]$ or for $\frac{- -1 \pm \sqrt{(-1)^2 - 4(18)(-4)}}{2(18)}$ oe	M2	FT <i>their</i> 3-term quadratic M1 for factors that give two correct terms when expanded or for correct discriminant or correct $\frac{-b}{2a}$ provided quadratic formula is in correct form
	$[0].5$ or $\frac{1}{2}$ final answer	B1	
6(a)(i)	$\frac{1}{3}$ oe	1	
6(a)(ii)	0	1	
6(a)(iii)	$\frac{1}{6}$ oe	1	
6(b)(i)	$\frac{1}{15}$ oe	2	M1 for $\frac{2}{6} \times \frac{1}{5}$ or equivalent method
6(b)(ii)	$\frac{4}{15}$ oe	3	M2 for $\frac{2}{6} \times \frac{1}{5} + \frac{3}{6} \times \frac{2}{5}$ or equivalent method or M1 for $\frac{2}{6} \times \frac{1}{5}$ oe seen or $\frac{3}{6} \times \frac{2}{5}$ oe seen
6(c)	$\frac{7}{18}$ oe	3	M2 for $\left(\frac{1}{6}\right)^2 + \left(\frac{2}{6}\right)^2 + \left(\frac{3}{6}\right)^2$ oe or M1 for one correct product seen or sample space with 14 correct pairs identified
7(a)	2, 4.5	2	B1 for each
7(b)	Correct graph 	4	B3 FT for 6 or 7 correct points FT <i>their</i> table or B2 FT for 4 or 5 correct points FT <i>their</i> table or B1 FT for 2 or 3 correct points FT <i>their</i> table

Question	Answer	Marks	Partial Marks
7(c)(i)	−0.5 to −0.4	1	
7(c)(ii)	$y = 1 - x$ ruled and −1.9 to −1.75	2	M1 for $[y =] 1 - x$ or $\left[x^2 + \frac{1}{x} =\right] 1 - x$ soi or B1 for −1.9 to −1.75
7(d)	Any integer ≥ 2	1	
8(a)	$[v =] 40$ $[w =] 80$ $[x =] 40$ $[y =] 100$ $[z =] 60$	5	B1 for each FT angle z as $140 - \text{their } w$
8(b)	24	3	M2 for $360 - 11x = 2 \times 2x$ oe or M1 for $360 - 11x$ seen or obtuse angle $KOL = 2 \times 2x$ oe
8(c)(i)	angle $ADX = \text{angle } BCX$ oe same segment oe angle $DAX = \text{angle } CBX$ oe same segment oe angle $AXD = BXC$ oe [vertically] opposite oe	M2	Accept in any order M1 for one correct pair with reason If 0 scored, SC1 for two correct pairs of equal angles identified with incorrect/no reasons
	corresponding angles are equal oe	A1	
8(c)(ii)(a)	8.75 or $8\frac{3}{4}$	2	M1 for $\frac{8}{10} = \frac{7}{DX}$ oe
8(c)(ii)(b)	81.8 or 81.78 to 81.79	4	M2 for $[\cos[BXC] =] \frac{5^2 + 7^2 - 8^2}{2 \times 5 \times 7}$ oe or M1 for $8^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \cos(\dots)$ oe A1 for $\frac{10}{70}$ oe

Question	Answer	Marks	Partial Marks
9(a)	315 or 314.5 to 315.0	6	M1 for $\tan 70 = \frac{\text{height}}{\frac{1}{2}(8-5)}$ oe or better seen M1dep for $\frac{1}{2}(8+5) \times \text{their height}$ or better seen dep on trig attempt for height M2 for $12 \times \frac{\frac{1}{2}(8-5)}{\cos 70}$ oe or better seen or M1 for $\frac{\frac{1}{2}(8-5)}{\cos 70}$ oe or better seen M1 for 8×12 oe isw and 5×12 oe isw
9(b)(i)	$8 - \frac{1}{2}(8-5)$ or $5 + \frac{1}{2}(8-5)$	M1	
9(b)(ii)	13.6 or 13.64 to 13.65	2	M1 for $12^2 + (6.5)^2$ oe
9(b)(iii)	16.8 or 16.9 or 16.79 to 16.91... nfw	2	M1 for identifying angle GAX from a diagram or from working or better
10(a)(i)	10	1	
10(a)(ii)	-19	1	FT 1 – 2 <i>their (a)(i)</i>
10(b)	$\frac{1-x}{2}$ oe final answer	2	M1 for $x = 1 - 2y$ or $y + 2x = 1$ or $\frac{y}{2} = \frac{1}{2} - x$ or $y - 1 = -2x$ or better
10(c)	$\frac{1}{2}$ oe	1	
10(d)	$4x^2 - 8x + 2$ final answer	4	M1 for $(1-2x)(1-2x) - (1-2(1-2x))$ or better B1 for $1 - 2x - 2x + 4x^2$ B1 for $-(1-2+4x)$ or better or $[+] 1 - 4x$ or for correct answer seen then spoiled
10(e)	x final answer	1	
10(f)	3125	1	
10(g)	25	1	
10(h)	-2	2	B1 for $\frac{1}{25}$ or 0.04
11(a)	A : -3 $17 - 4n$ oe	3	B1 for -3 B2 for $17 - 4n$ oe or B1 for $k - 4n$ oe or $17 - pn$ oe, $p \neq 0$

Question	Answer	Marks	Partial Marks
	B : 124 $n^3 - 1$ oe	3	B1 for 124 B2 for $n^3 - 1$ oe or B1 for any cubic
	C : $\frac{11}{128}$ $\frac{n+6}{2^{n+2}}$ oe	4	B1 for $\frac{11}{128}$ B3 for $\frac{n+6}{2^{n+2}}$ oe or B2 for 2^{n+2} oe seen or B1 for 2^k oe or $n + 6$ seen
11(b)	$\frac{p+1}{2q}$ oe	2	B1 for $p + 1$ or $2q$ oe